

Mathew Stevens

Full-Stack Software Engineer | AI Agent Infra | Real-Time Systems | SDKs

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Full-stack software engineer who ships production software solo and operates it under real load: two live apps across web, iOS, and Android with 2,250+ users, 66 reviews, and app-store releases. Also builds the layer underneath, from deterministic execution and exactly-once external actions to evaluation tooling that catches silent agent failures.

EXPERIENCE

Independent Full-Stack Engineer / Product & Systems

2025 - Present

- Built and deployed Agentic, an AI-agent authorization layer across web, mobile, CLI, and desktop, reaching 1000+ users in under 1 month with 40 reviews and a 4.2-star rating.
- Built and launched SolPulse, a production trading platform across web, Android, and iOS, with an execution engine sustaining 10ms monitoring across up to 1,200 concurrent transactions.
- Built AI-agent reliability/eval systems for exactly-once external actions and crash/recovery conformance; authored the original Terminal-Bench durable-outbox task where 6/6 frontier-agent trials failed clean reward-0.
- Invited by AI Fund (Andrew Ng) as one of 12 engineers to build against a B2B sales problem posed by a Fortune 100 OEM; delivered ConfigPilot, a decision engine gating LLM recommendations behind deterministic margin, inventory, and policy checks.
- Built Capability Record Replay, a model-to-deterministic-replay runtime where one LLM discovery run becomes a typed, approved capability that later executes with no model in the decision path; conformance suite killed 9/9 weakened replay engines.
- Built cross-platform SDKs with native Android/iOS bridges and cryptographic signing across Unreal, Cocos, Godot, Unity, and Capacitor; contributed four merged upstream PRs to the Godot and Unity adapters.

Analytics Specialist, Socket Mobile, Inc.

Apr 2020 - Jul 2023

Fremont, CA

- Owned web analytics end to end in Google Analytics and Google Tag Manager: data collection, tagging, QA, A/B and multivariate testing, dashboards, and reporting for marketing and leadership.
- Partnered with the web team on the analysis behind a 25% year-over-year sales increase, turning traffic and conversion data into channel and campaign decisions leadership acted on.

SELECTED PROJECTS

Agentic - AI Agent Signing Platform

Web, CLI, Desktop, App Store

- Built an authorization layer where AI agents propose actions and users approve them on their own devices; private keys never leave user custody.
- Shipped across web, mobile, CLI, desktop, MCP tools, SDK surfaces, and app-store releases behind one non-custodial approval boundary.
- Implemented shared WalletBackend plus A2A AgentCard, AP2/ACP/MPP payment-adaptor flows, MCP/Vercel AI tools, approval polling, simulation, and sign-and-send UX.

SolPulse - Trading Platform

React, Node, PostgreSQL, Redis

- Built solo full-stack trading product with live website, App Store, Google Play, and browser-visible release proof.
- Implemented execution engine with DCA, limit orders, conditional exits, multi-venue routing, dynamic fee optimization, and real-time strategy logs.
- Built risk and fraud detection across 35 scoring signals: counterparty identity checks, anomaly and large-holder detection, max daily loss limits, and per-trade exposure caps.

Agent Reliability & Eval Integrity Systems

Python, TypeScript, Docker, Terminal-Bench

- Built Agent Eval Foundry: 25 coding-agent benchmark packages and 170 Claude/Codex attempts; 12 task histories meet the failure target after audit (nine at 6/6 failures, three at 5/6), with preserved evidence and two-sided grader validation.
- Built the original Klavis Terminal-Bench durable-outbox task plus its reference implementation, which passes 267/267 conformance checks.
- Built Durable Agent Outbox and Crashpoint to measure exactly-once external side effects, UNKNOWN outcomes, receipt reconciliation, revocation, and crash/recovery across LangGraph, Temporal, and DBOS.
- Built Cheat-Oracle and Judge-Artifact audits showing detector undercount and grader artifact in public agent benchmarks, with tamper-evident receipts, controls, and disclosure-ready reports.

Capability Record Replay - Agent Automation Runtime

TypeScript, Playwright, Anthropic, CLI

- Built system where an LLM drives a legacy back-office UI once, then synthesis emits a typed, versioned capability artifact replayed without a model in the decision path.
- Implemented browser and terminal surfaces, deterministic interpreter, policy chokepoint, signed approvals, typed business outcomes, recovery/failure taxonomy, evidence journals, and redaction canaries.
- Verified a live Claude Opus discovery run and model-free replay across 25 hostile fixture scenarios.

Cross-Platform SDKs and Engine Integrations

Unreal, Cocos, Godot, Unity, iOS

- Built Unreal Engine 5 mobile wallet plugin with Kotlin clientlib, C++ JNI bridge, Blueprint UFUNCTION surface, auth caching, SIWS, and sign-and-send flows.
- Built the first Cocos Creator Mobile Wallet Adapter SDK from scratch: TypeScript API, native Android bridge, ECDH/AES-GCM wallet association, auth caching, and Token Duel demo game.
- Delivered Godot MWA reconnect architecture, SIWS wallet-proof APIs, persistent auth-token cache, Unity WebGL UX fix, and iOS-native Swift/CryptoKit adapter flows.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Node.js, Swift, C#, C++, GDScript, Kotlin, SQL, Python

Frontend: React, Vite, Zustand, Tailwind, Radix UI, Capacitor, SwiftUI

Backend: Express, FastAPI, Pydantic, PostgreSQL, Prisma, Redis, WebSockets, SSE, REST APIs

Security / Transaction Systems: web3.js, Mobile Wallet Adapter, Wallet Standard, Seed Vault, Jupiter, Anchor, SPL tokens, transaction signing

Platforms: Android, iOS, Unity, Unreal Engine, Godot, Cocos Creator

Systems / AI Infra: Docker, CI/CD, Terminal-Bench, Harbor, conformance suites, mutant tests, deterministic replay, policy gates, eval-integrity audits

LAUNCH METRICS

1000+

SolPulse users

4.3 / 26

SolPulse rating/reviews

1250+

Agentic users

4.2 / 40

Agentic rating/reviews

2

Store-listed apps

6

Merged upstream PRs

OPEN-SOURCE PROOF

- GitHub portfolio: github.com/mstevens843
- Agent Eval Foundry: 25 packages, 170 attempts; audited results, grader controls and preserved evidence.
- Godot Mobile Wallet Adapter PRs: #449 reconnect/capabilities/sign-send bridge; #453 SIWS wallet-proof APIs; #454 auth-token cache.
- Unity Wallet Adapter merged PR: #275 WebGL single-wallet auto-select UX fix.
- Roboflow merged PRs: #114 oriented-box annotation renderer (supervision-js); #3001 builder link discovery and activation fix (inference).
- Project proof: capability-record-replay, durable-agent-outbox, crashpoint, cheat-oracle, judge-artifact, agent-context-containment, model-regression-sentinel, toolcall-risk-classifier, ConfigPilot, Klavis TB3, TxShield.

UPSTREAM IMPACT

- Crashpoint / DBOS:** Supplied Postgres crash/recovery evidence supporting merged workflow-identity fix #838.
- Crashpoint / LangGraph:** Reported pre-checkpoint gap #8764; reproduction adopted in AgentCLI's merged fixture #160.
- TrueForge / TrueFoundry:** Reported restart-recovery semantics gap #773 for MCP effects that commit before tool.response, backed by a 180-trial crash-boundary evidence matrix.
- Judge-Artifact / Inspect:** Contributed 3,986-transcript grader analysis and parser regressions to #2108 / #2310.

EDUCATION

San Francisco State University

B.S., Business Information Systems, 2018

Springboard

Software Engineering Certificate, 2025

Full-stack curriculum covering JavaScript, React, Node.js, Express, PostgreSQL, REST APIs, auth, and testing.

FOCUS AREAS

Full-Stack Products

AI Agent Infra

Eval Integrity

Real-Time Systems

Mobile Apps

SDK Engineering

Best-fit roles: full-stack product engineering, backend/API systems, AI-enabled systems, reliability/eval tooling, SDKs, mobile apps, and real-time infrastructure.